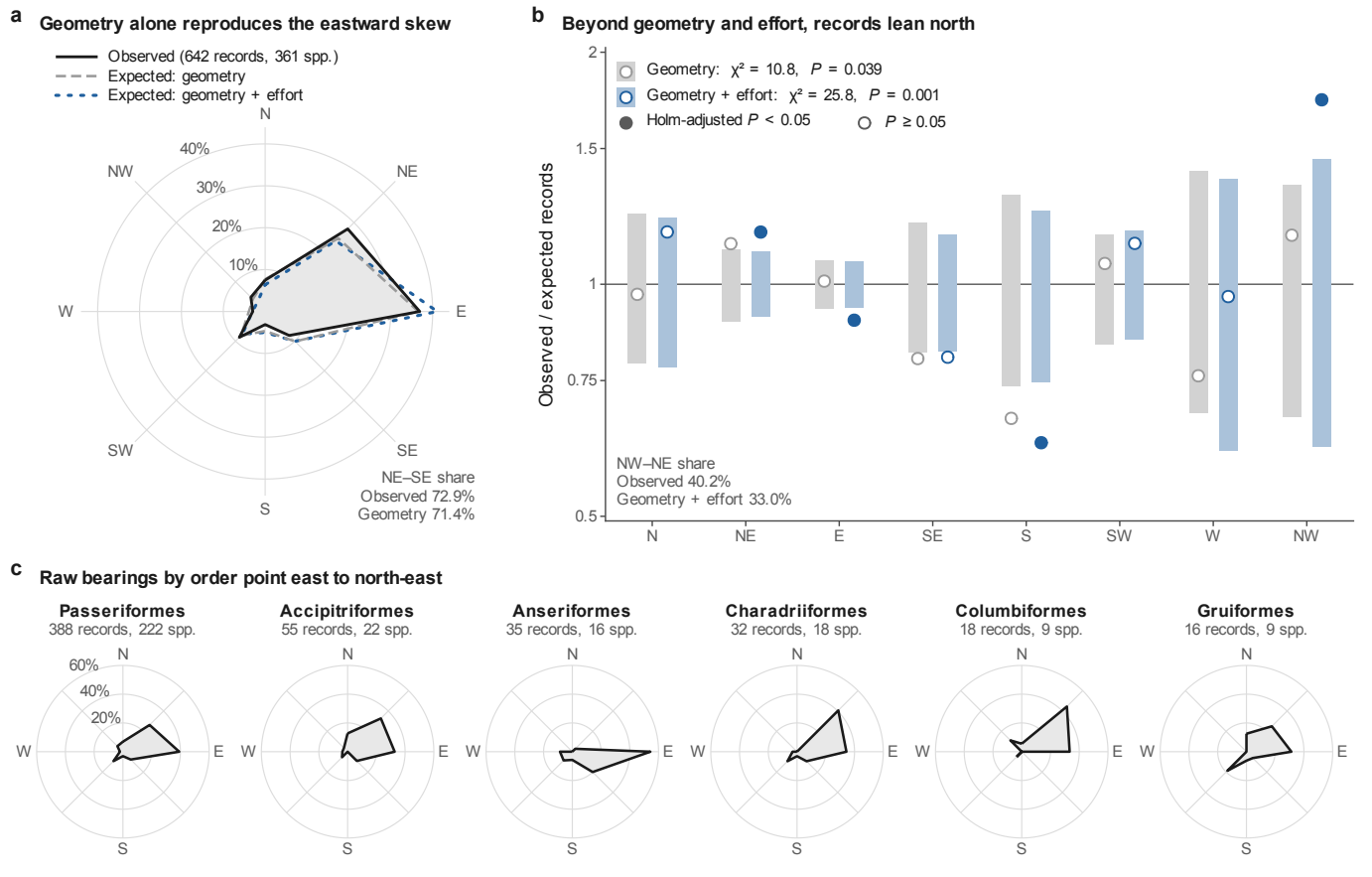

a
Geometry alone reproduces the eastward skew
Observed (642 records, 361 spp.)
Expected: geometry
Expected: geometry + effort
N
NE
E
SE
S
SW
W
NW
40%
30%
20%
10%
NE–SE share
Observed 72.9%
Geometry 71.4%
b
Beyond geometry and effort, records lean north
Geometry: χ² = 10.8, P = 0.039
Geometry + effort: χ² = 25.8, P = 0.001
Holm-adjusted P < 0.05
P ≥ 0.05
Observed / expected records
2
1.5
1
0.75
0.5
NW–NE share
Observed 40.2%
Geometry + effort 33.0%
N
NE
E
SE
S
SW
W
NW
c
Raw bearings by order point east to north-east
Passeriformes
388 records, 222 spp.
Accipitriformes
55 records, 22 spp.
Anseriformes
35 records, 16 spp.
Charadriiformes
32 records, 18 spp.
Columbiformes
18 records, 9 spp.
Gruiformes
16 records, 9 spp.
60%
40%
20%